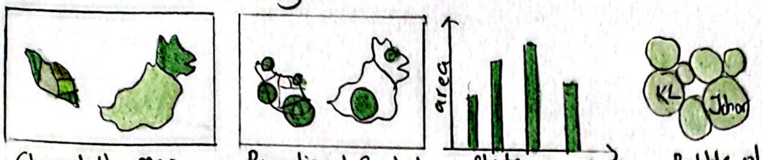
SHEET 1

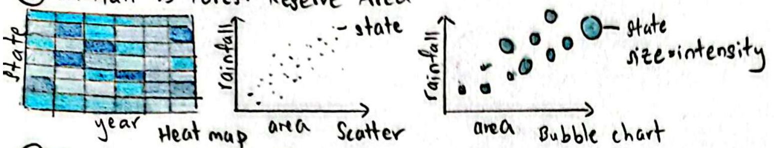
IDEAS

① Forest Reserve Coverage



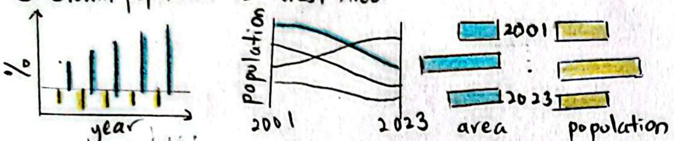
Choropleth map · Proportional Symbol · state Bar chart · Bubble plot

② Rainfall vs Forest Reserve Area



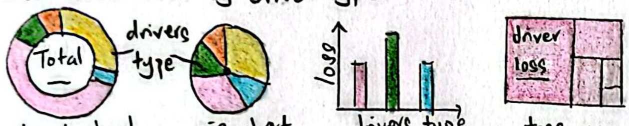
Heat map · Scatter · Bubble chart

③ Growth population vs Forest Area



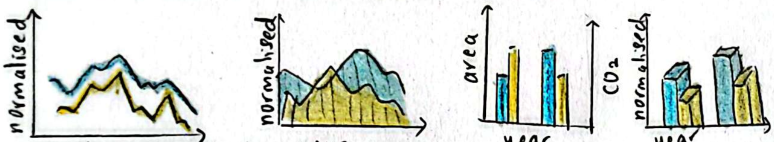
Bar chart · slope chart · butterfly chart

④ Forest Cover Loss by Driver type



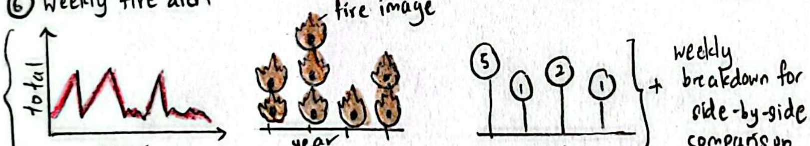
donut chart · pie chart · drivers type bar chart · tree map

⑤ Forest Cover Loss vs CO_2 emissions



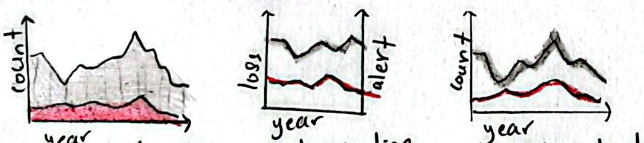
Multi-line chart · stacked area · dual-axis grouped bar · 3D grouped bar

⑥ Weekly fire alert



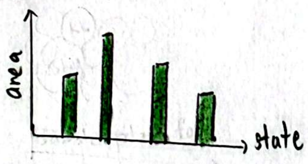
line chart · isotype chart · lollipop chart + weekly breakdown for side-by-side comparison

⑦ Fire alert vs tree cover loss

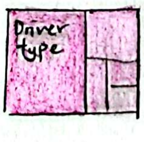


stacked area · dual-axis line · multi-line chart

FILTER



Forest Reserve Coverage
- loss information on regional location on map
- not user-friendly for people unfamilar with malaysia regional location.



Tree map
- small values hard to see
- difficult to compare size precisely
- human have lower ability in area comparison

3D grouped bar
- high data-ink ratio
- extra dimension don't bring extra information
- misintepretation

CATEGORIZE

① Forest Reserve Coverage

↓

② Rainfall vs Forest Reserve Area

↓

③ Growth population vs Forest Area

↓

④ Forest Cover Loss by Driver Type

↓

⑤ Forest Cover Loss vs CO_2 emissions

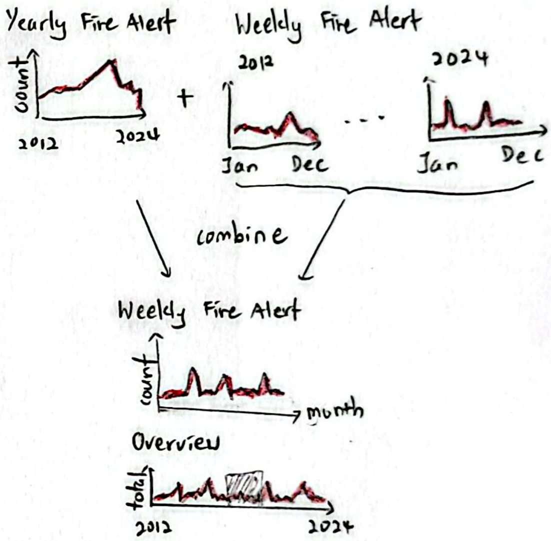
↓

⑥ Weekly fire alert

↓

⑦ Fire alert vs Tree cover loss

COMBINE & REFINE



Without combining, we will be showing small multiples of line chart representing every year — weekly trend, that is not effective as its too large number of charts.

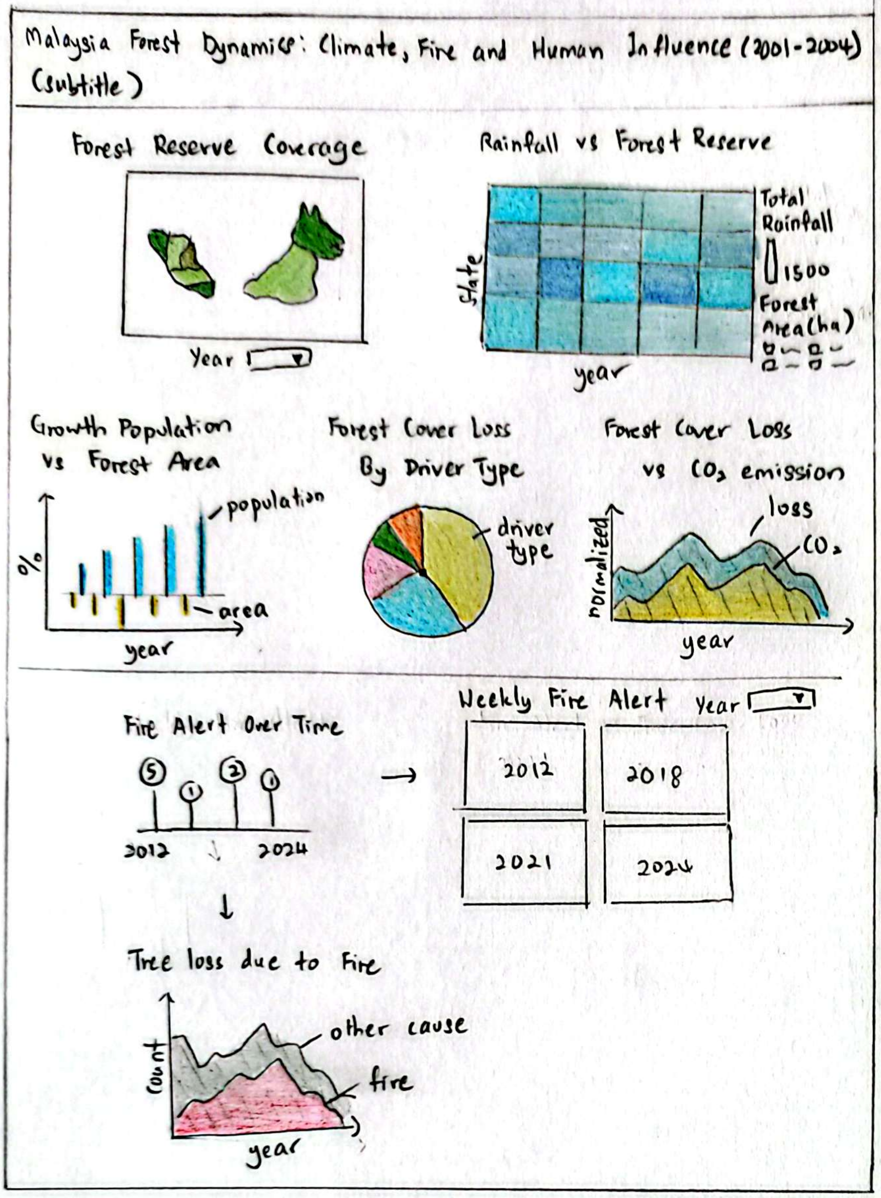
- After combining, save up more space, facilitate advanced user interaction

QUESTION

1. Is all the visualization idioms doable?
2. Is the data-ink ratio high?
3. Will our client understand our idiom easily at first sight?
4. Is the colour choice appropriate?

SHEET 2

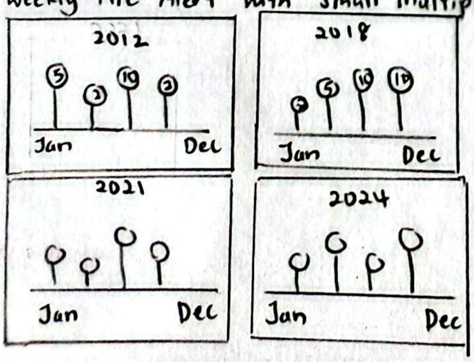
LAYOUT



FOCUS

Forest Reserve Coverage

Weekly Fire Alert with Small Multiples Interaction



User can directly perform side-by-side comparison without year slider / filtering.

Title: Simple Row Layout Dashboard Idea
Author: Goh Jia Xuan
Sheet: 2
Date: 13/10/2025
Task: Design row layout dashboard for 7 study topics

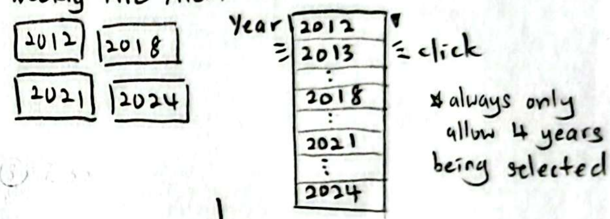
OPERATIONS

① Year filter

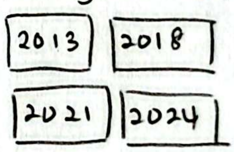
For choropleth map, user can choose specific year to visualize, otherwise, it will only show the mean across all years.

For weekly fire alert, by default it will always show 2012, 2018, 2021, 2024 visualization. User can later choose other year to visualize via dropdown menu.

Weekly Fire Alert



Weekly Fire Alert



Now, 2012 Weekly Fire Alert is replaced with 2013 Weekly Fire Alert.

DISCUSSIONS

Pros:

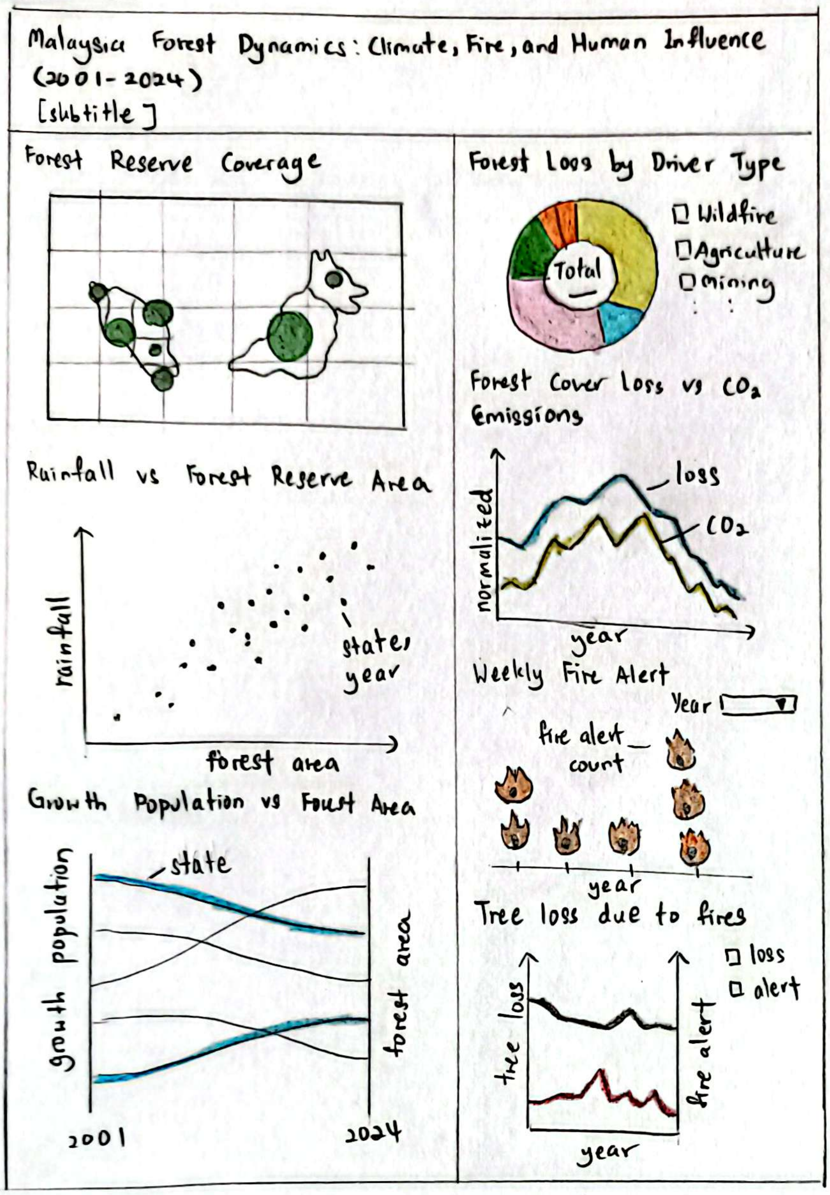
1. Additional layers like ocean, foreign countries allows better visual context and spatial grounding.
2. Map looks more professional, natural.
3. Less repeating idiom shows more variety.

Cons:

1. Inconsistent row layout, less tidy.
   ↳ e.g. 1st row has 2 idioms, 2nd row has 3 idioms, 3rd row has 2 idioms, last row have 1 idiom
2. Empty space at bottom right corner.
3. Horizontal scrolling may be required.

SHEET 3

LAYOUT



Title: Simple Column Layout Dashboard Idea

Author: Goh Jia Xuan

Sheet: 3

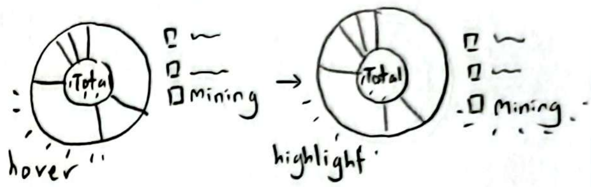
Date: 13/10/2025

Task: Design column layout dashboard for 7 study topics

OPERATIONS

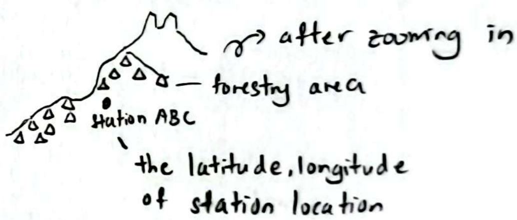
① Highlighters

- user hover on any driver type, the driver will be highlighted; others will be in shaded colour



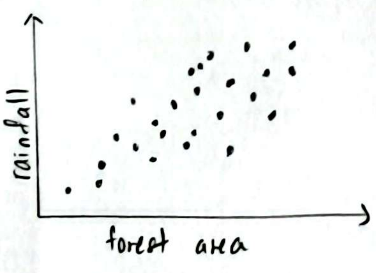
② Zoom in, zoom out

- When user hovering the proportional symbol map, if user scroll up their mouse, the map will zoom in to show the bubble information. i.e. the station location, the forestry area



FOCUS

Sightline: top to bottom.

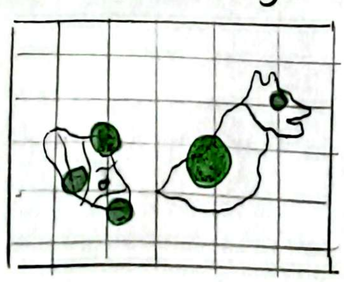


states
- ☐ Johor
- ☐ KL
- ☐ Melaka

all states in Malaysia

- colour hue encodes different state.
- each point encode the year

Forest Reserve Coverage



→ graticule 1° or 5°

- the bubble size indicates coverage area
- larger bubble shows larger forestry area.

DISCUSSIONS

Pros:

1. Simple, clear layout, easy to understand.
2. Zooming in and out interaction allows advanced user interaction.
3. Consistent sightline.
   ↳ top → bottom.

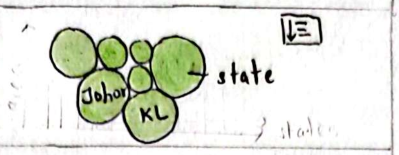
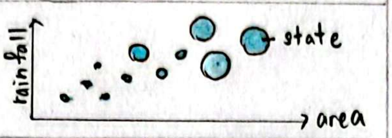
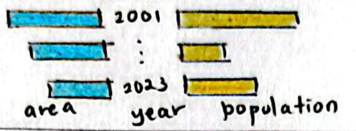
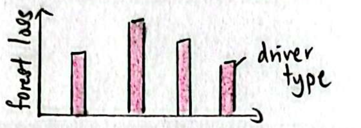
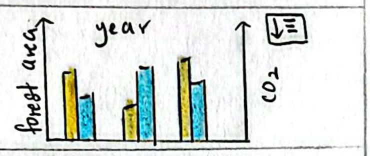
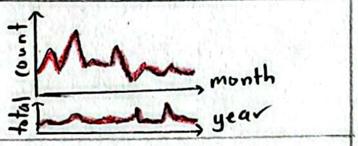
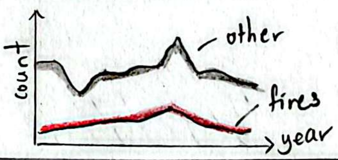
Cons:

1. Proportional symbol map may have overlapping issue in term of the bubble size.
2. Proportional symbol map will need the latitude, longitude data to plot the bubble at the state. i.e. the station
3. May require horizontal scrolling.

SHEET 4

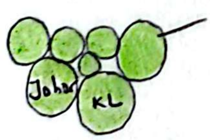
LAYOUT

Malaysia Forest Dynamics:
Climate, Fire and Human Influence (2001-2024)
(subtitle)

Forest Reserve Coverage [description 1]	
Rainfall vs Forest Reserve Area [description 2]	
Growth population vs forest area [description 3]	
Forest loss by driver type [description 4]	
Forest cover loss vs CO_2 emission [description 5]	
Weekly fire alert [description 6]	
Tree loss due to fires [description 7]	

FOCUS

Forest Reserve Coverage

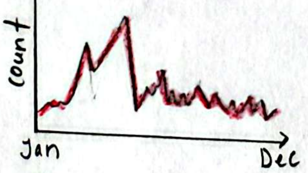


- each bubble represent a state.
- bubble size indicate the coverage area in kilometre square, km².

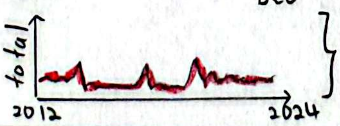
size = forest area
○<10K ◎10-20K ●>20K — a legend indicate the bubble size labelling

- the larger the bubble, the larger the forest area for that specific state.

Weekly Fire Alert



detail section: the breakdown into every week. When user hover on the point, the specific date will be shown.

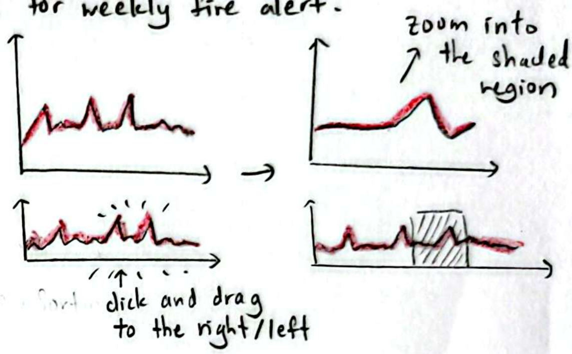


Overview section: user view the annual trend

Title: Design storytelling narrative layout
Author: Goh Jia Xuan
Sheet: 4
Date: 13/10/2025
Task: Design narrative layout dashboard for 7 study topics

OPERATIONS

1. Overview + Detail
   - user click and drag the overview section to view the specific range of year for weekly fire alert.



2. Sorting
   - user can sort based on forest area / CO_2, in ascending or descending order

DISCUSSIONS

Pros:

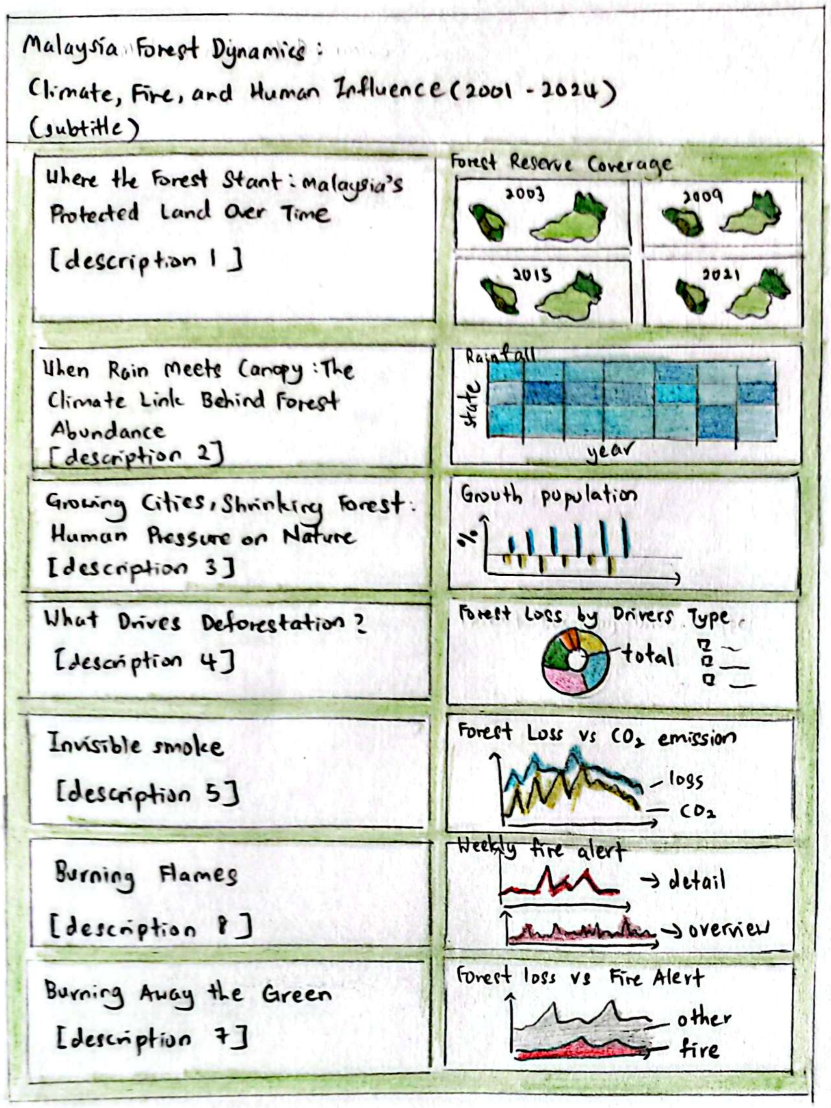
1. Sufficient space for text annotation / description.
2. Consistent sightline, structured layout increase readability.
3. Advanced interaction improves user interaction. i.e. Overview + Detail.

Cons:

1. Bubble plot will loss regional location for each state on the map.
2. May be difficult to have fixed width and height for each idiom.
3. Extra attention needed in ensuring text annotation does not exceed the height of the idiom too much as it may cause unnecessary white space on the idiom as well.

SHEET 5

LAYOUT



Title : Design Final Implementation Design

Author : Goh Jia Xuan

Sheet : 5

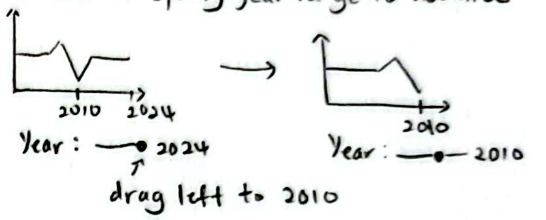
Task : Design final layout to visualize 7 study topics.

Date : 13/10/2025

OPERATIONS

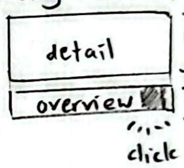
① Year Slider

for forest cover loss vs CO_2 emissions and tree cover loss composition, a year slider is used to allow user to specify year range to visualize



② Overview + Detail

Weekly Fire Alert

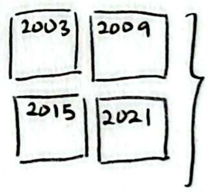


} zoom into the selected time window to see weekly pattern

} show annual fire alert trend

③ Small multiples

Forest Reserve Coverage

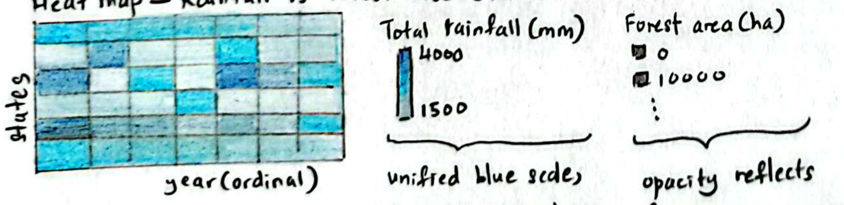


- show choropleth map across different year
- visually compare changes over time without needing interaction

FOCUS

The visual sequence from top → bottom, left → right.

Heat map — Rainfall vs Forest Reserve Area



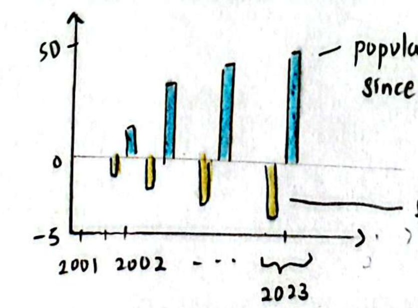
unified blue scale, where intensity shows rainfall

opacity reflects forest reserve area

Dark + opaque blue → abundant rain, large forest

Faint + translucent blue → limited rainfall, less forest

Bar chart — Indexed Change of Population and Forest Area



— population changes for every year since 2021 (%)

— forest area changes for every year since 2021 (%)

DETAIL

Dependencies & Tools :

- Python : data cleaning, aggregation, formatting
- Jupyter : environment for data preparation
- Vega-Lite : create idiom
- CSS : content alignment, size, orientation

Estimated Time & Effort :

- 2 days : data preparation
- 1 day : line chart, stacked area chart
- 1 day : choropleth map, multiline chart
- 1 day : bar, heat map, donut chart
- 1 day : dashboard, tooltip, annotation
- Total : 6-7 days

Description :

- sum, normalization, average value over time is required.